

GABRIEL PUJOL

Senior Software Developer - Technical Lead - Future Engineering Manager

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PROFESSIONAL SUMMARY

- Senior Software Developer with 10+ years of experience delivering solutions across Healthcare, Oil & Mining, and Geostatistics industries.
- Proficient in Python and the .NET ecosystem, with extensive hands-on experience in both Desktop and Backend development.
- Proven ability to lead and mentor technical teams, driving Agile practices through daily stand-ups, code reviews, and one-on-one coaching.
- Recognized for clear communication and fluency in Spanish, English, French, and Portuguese.
- Passionate about Clear Code, Algorithms, Metaheuristics, and applied Optimization, with a strong focus on designing, analyzing, and deploying robust, production-grade systems that deliver measurable business value.

EDUCATION

PhD in Computer Science, Optimization (*unfinished*) 2014 – 2018
Pontifical Catholic University of Rio de Janeiro, Rio de Janeiro, Brazil.

- Proposed a novel Geostatistical Simulation algorithm based on Multiple-Point Statistics, and conditioned by Seismic, Soft, and Hard Data.
- The research extends and generalizes the algorithm proposed in the paper “LSHSIM: A Locality Sensitive Hashing based method for multiple-point geostatistics”.
- Keywords: Geostatistical Simulation, Multiple-Point Geostatistics, C++, Python.

Master's degree in Mathematics 2012 – 2015
University of Havana, Havana, Cuba.

- Designed and developed a program to optimize the traffic flow of complex street intersections.
- This work combined Simulation Theory and Metaheuristics and resulted in a patent (see below).
- Keywords: Multi-objective optimization, NSGA-II, Mathematical modelling, Matlab, Simulation.

Diploma in Neurosciences 2011 – 2012
Cuban Neurosciences Center

- Gained an overview of the state-of-the-art of Neuroscience Research, with emphasis on Medical Imaging.
- To conclude the degree, a seminar was prepared on the [Blue Brain Project](#).
- Keywords: Medical Imaging, Matlab, Statistics.

Bachelor's degree in Computer Science 2006 – 2011
University of Havana, Havana, Cuba.

- Designed and implemented an efficient storage format for large volumes of vector geographic data.
- This work was added to a commercial Geographic Information System (GIS), improving localized queries.
- Keywords: C#, GIS, Mono, Mono Migration Analyzer (MOMA).

PROFESSIONAL EXPERIENCE

Software Developer ([AlayaCare](#))

March 2025 – Present

- Collaborated with Product Owner and Project Manager to analyze and define evolving client requirements.
- Owned and delivered key application modules supporting those requirements.
- Contributed to code reviews and deployed into CI/CD pipelines to ensure high-quality, maintainable releases.
- Worked with cross-functional teams to resolve production issues and improve system performance.
- Keywords: Python, Docker, FastAPI, SQLAlchemy, PostgreSQL.

Senior Software Developer ([Graitec Group](#))

December 2024 – February 2025

- Co-hosted daily stand-ups and weekly progress reviews, supporting team alignment and delivery.
- Participated in code reviews, integrations, and deployments to ensure quality and stability.
- Contributed to the development of modules and algorithms with a focus on performance, maintainability, and compatibility.
- Keywords: C#, .NET platform, Revit, AutoCAD, Plugin Architecture.

Sales and Business Development Associate ([MVA Power Inc.](#))

June 2024 – December 2024

- Engaged with new and existing clients to gather requirements and deliver tailored solutions.
- Led negotiations, aligning client needs with company capabilities to close deals.
- Strengthened stakeholder communication and cross-department collaboration skills.
- Contributed to project planning and execution, defining scope, timelines, and deliverables.
- Keywords: Requirement Analysis, Project Management, Sales, Customer Relations.

Software Developer ([ALS Goldspot Discoveries Ltd.](#))

June 2023 – May 2024

- Contributed to the design and development of a cloud-based platform on AWS for remote geological analysis and data science workflows.
- Participated in code reviews to enforce best practices, maintainability, and code quality.
- Led performance optimization efforts through service parallelization, cutting running time by over 50%, which is especially sensitive on processing large data volumes.
- Developed an automated code generation tool to streamline configuration updates and reduce manual effort.
- Keywords: Python, Pandas, Scikit-Learn, Matplotlib, Machine Learning tools.

Software Developer ([Géotic Inc.](#))

October 2018 – January 2024

- Contributed to the development of the Géotic software suite: GeoticLog, GeoticMine, and GeoticField.
- Designed and implemented scalable data storage solutions to ensure data integrity and accessibility.
- Improved portability and usability of field tools to support geological surveys in remote environments.
- Keywords: C#, .NET platform, Microsoft SQL Server, Microsoft Office.

Software Developer & Researcher ([PUC-Rio, Brazil](#))

August 2014 – September 2018

- Designed and delivered a scalable geostatistical simulation framework based on Multiple-Point Statistics, integrating seismic, soft, and hard data.
- Collaborated with domain experts to optimize performance for large datasets and improve exploration and risk assessment decisions.
- Keywords: C++, Python, Optimization, Geostatistical Simulation.

- Developed Optima Pesquisa, an Android-based tool used in primary schools to screen children for neurodevelopmental deviations.
- Designed and delivered Neuronic Optima Pesquisa, a multi-platform system supporting regional programs for early detection of neurodevelopmental risks.
- Keywords: Java, Android, Eclipse.

SKILLS

- Programming Languages: C#, Python, C++, Java.
- Databases: PostgreSQL, Microsoft SQL Server, SQLite,
- Frameworks & Tools: .NET Suite, AWS, GitHub, Cursor, Docker, FastAPI, Pandas, Scikit-Learn, Matplotlib, SQLAlchemy, Matlab.
- Soft Skills: Leadership, technical mentoring, cross-team communication, analytical problem-solving, code review and quality standards, requirement analysis, project management, sales, customer relations.
- Languages: Spanish (native), English (fluent), Portuguese (fluent), French (fluent).

PUBLICATIONS & PATENTS

- Moura, Pedro, Eduardo Laber, Hélio Lopes, Daniel Mesejo, Lucas Pavanelli, João Jardim, Francisco Thiesen, and **Gabriel Pujol**. “LSHSIM: a locality sensitive hashing based method for multiple-point geostatistics.” *Computers & Geosciences* 107 (2017): 49-60.
- Guilarte, Orlando F., **Gabriel S. Pujol Fariña**, Sira M. Allende Alonso, and Gemayqzel Bouza Allende. “Herramienta computacional para sugerir y validar políticas de luces de semáforos.” In *XIII Simposio Argentino de Investigación Operativa (SIO)-JAIIO 44* (Rosario, 2015). 2015.
- Fonseca Guilarte, Orlando, **Gabriel Santiago Pujol Fariña**, Sira Allende Alonso, and Gemayqzel Bouza Allende. “Una herramienta para simular y analizar el flujo de tráfico en un cruce regulado por semáforos con tiempos de ciclos variables.” *Revista Cubana de Ciencias Informáticas* 8 (2014): 29-40.
- Herramienta Computacional Para Sugerir y Validar Políticas de Luces de Semáforos (SUVASEM 3.0), Patent No. 0243-01-2018, Centro Nacional de Derecho de Autor (CENDA), Havana, Cuba.

INTERESTS

- Facing new challenges and learning new technologies.
- Writing simple and extensible code to solve real-world problems.
- Being up to date with the development of cutting-edge technology.
- Putting together my experiences into stories that I can tell, so others may learn them in an interesting way.
- In my free time, I like to play guitar, sing to my close ones, cook, and play video games.